

Linear Statistical Models

B. Statistical Data Science 2nd Year Indian Statistical Institute

Class Test - 1

27 February 2026

Total Marks: 10

Name:

Roll Number:

Instructions

- Answer as much as you can. The maximum you can score is 10.
- Only write the answer in the space provided or tick the correct option. Any additional writing will not be graded. Rough works should be done on spare pages.

Question 1. Which of the following are linear statistical models?

[4 × 0.5 marks]

The model needs to be linear in the parameters.

☒ (A) $E(y) = \alpha + \beta x + \gamma e^{5x^2}$

☒ (B) $E(y) = \beta_{11}x_1^2 + \beta_{12}x_1x_2 + \beta_{22}x_2^2$

☐ (C) $E(y) = (\beta_1x_1 + \beta_2x_2)^2$

☒ (D) $E(y) = \begin{cases} \alpha & \text{if } x < 50, \\ \alpha + \beta & \text{if } 50 \leq x < 75, \\ \beta + \gamma & \text{if } x > 100. \end{cases}$ 75

Question 2. Consider the linear regression model $y = X\theta + \epsilon$, where X is an $n \times d$ matrix and $\theta \in \mathbb{R}^d$. Which of the following conditions ensures that the normal equations $X^T X \theta = X^T y$ have a unique solution?

[0.5 marks]

☐ (A) $\text{rank}(X) = \text{rank}(X^T X)$

☐ (B) $\text{rank}(X) = n$

☒ (C) $\text{rank}(X) = d$

☐ (D) $\text{rank}(X^T X) = n$

X needs to have full col. rank.
($\Rightarrow$) $X^T X$ needs to be invertible

Question 3. Let P be an orthogonal projection matrix onto a proper subspace of $\mathbb{R}^n$. Which of the following is/are FALSE?

[1 mark]

(A) $(I - P)$ is symmetric and idempotent.

(B) $(I - P)$ is positive semi-definite.

(C) P and $(I - P)$ are orthogonal to each other.

☒ (D) $(I - P)$ is positive definite.

Question 4. Let P be an $n \times n$ orthogonal projection matrix. Identify the correct statement(s) from the following.

[1 mark]

(A) $P^2 v = v \quad \forall v \in \mathbb{R}^n$

☒ (B) $P^2 v = P v \quad \forall v \in \mathbb{R}^n$ — By definition

(C) $P^3 v = v \quad \forall v \in \mathbb{R}^n$

☒ (D) $P^3 v = P v \quad \forall v \in \mathbb{R}^n$ — $P^3 v = P P^2 v = P P v = P^2 v = P v$

Question 5. Let X be an $n \times d$ matrix with $\text{rank}(X) = d$. Which of the following is the orthogonal projection matrix onto $\mathcal{C}(X)$? [0.5 marks]

- (A) XX^T (B) $(X^T X)^{-1}$ (C) $X(X^T X)^{-1}X^T$ (D) $I - X(X^T X)^{-1}X^T$

Question 6. Based on the data $(x_1, y_1), \dots, (x_n, y_n)$, two practitioners are interested in fitting the model $y = \alpha + \beta x + \epsilon$. The first practitioner estimates α and β by the least-squares method. The second practitioner also uses the method of least-squares, but after mean-centering x and y . Find the correct statement about the estimates obtained by the two practitioners. [1 mark]

(A) Estimates of both the parameters are the same.

(B) The estimated intercepts are the same, but the estimated slopes may be different.

(C) The estimated slopes are the same, but the estimated intercepts may be different.

(D) Estimates of both the parameters are different.

$$\hat{\beta} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

does not change due to mean centering

$$\hat{\alpha} = \bar{y} - \hat{\beta} \bar{x}$$

becomes

$$\hat{\alpha} = 0$$

Question 7. Consider the simple linear regression model $y_i = \alpha + \beta x_i + \epsilon_i$, $i = 1, \dots, n$. The fitted least-squares regression line based on 8 pairs of observations $(x_1, y_1), \dots, (x_8, y_8)$ is $\hat{y} = 2 + 0.5x$. Also, $\sum_{i=1}^8 (x_i - \bar{x})^2 = 8$ and $\sum_{i=1}^8 (y_i - \bar{y})^2 = 4$. Which of the following is the value of the coefficient of determination R^2 for this model? [1.5 marks]

$\hat{\beta} = 0.5 \Rightarrow \sum (x_i - \bar{x})(y_i - \bar{y}) = \hat{\beta} \sum (x_i - \bar{x})^2 = 0.5 \times 8 = 4$

$R^2 = r^2 = \frac{(\sum (x_i - \bar{x})(y_i - \bar{y}))^2}{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2} = \frac{4^2}{8 \times 4} = \frac{1}{2}$

(A) 0.25 (B) 0.40 (C) 0.50 (D) 0.75

Question 8. Let $\mathbf{v}_1 = (1, 2, -1)^T$ and $\mathbf{v}_2 = (-1, 1, 0)^T$. Determine scalars a_1 and a_2 such that $a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2$ is the orthogonal projection of $(1, 2, 3)^T$ onto the span of $\mathbf{v}_1$ and $\mathbf{v}_2$. [1.5 marks]

Answer: You are asked to find the projection of $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ onto $\mathcal{C}\left(\begin{bmatrix} 1 & -1 \\ 2 & 1 \\ -1 & 0 \end{bmatrix}\right)$

So, $(a_1, a_2) = (X^T X)^{-1} X^T y$, where $X = \begin{bmatrix} 1 & -1 \\ 2 & 1 \\ -1 & 0 \end{bmatrix}$, $y = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$

You can also obtain (a_1, a_2) by minimizing $\left\| \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} - a_1 \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} - a_2 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \right\|^2$ w.r.t. a_1, a_2

Alternatively, you can use the fact that the residual $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} - a_1 \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} - a_2 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$ must be orthogonal to $\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$.

Question 9. Consider a linear regression setup: $y_i = \alpha + \beta x_i + \epsilon_i$, $i = 1, \dots, 40$, with $\epsilon_i \stackrel{\text{i.i.d.}}{\sim} N(0, 4)$. We also know that $\sum_{i=1}^{50} x_i^2 = 60$ and $\sum_{i=1}^n x_i = 20$. Let $(\hat{\alpha}, \hat{\beta})$ be the OLS estimator of (α, β) . Determine $\text{Var}(\hat{\alpha})$ and $\text{Var}(\hat{\beta})$. [1.5 + 1.5 marks]

Answer:

$$\text{Var}(\hat{\alpha}) = \sigma^2 \left\{ \frac{1}{n} + \frac{\bar{x}^2}{\sum (x_i - \bar{x})^2} \right\}, \quad \text{Var}(\hat{\beta}) = \frac{\sigma^2}{\sum (x_i - \bar{x})^2}$$

$n = 40, \sigma^2 = 4, \bar{x} = \frac{20}{40} = \frac{1}{2}$

$\sum (x_i - \bar{x})^2 = \sum x_i^2 - n \bar{x}^2 = 60 - 40 \times \frac{1}{4} = 50$

$\text{Var}(\hat{\alpha}) = \frac{3}{25}, \quad \text{Var}(\hat{\beta}) = \frac{2}{25}$